

Fear of Temperature: Stratified Lexical Expansion, Voice Mapping, and Semantic-Relational Pre-Validation — Round 1B

Executive Summary and Methodological Transition

Round 1B substantially strengthens the lexical evidence base for the six fixed anchors—1842, 1938, 1988, 2006–2007, 2015, and 2022—by treating **speaker/voice, source genre, expression mode, and research provenance as analytically separate dimensions**. That distinction is indispensable: a scientist quoted in a newspaper remains a scientific voice; a survey respondent remains a lay/public voice even when the quotation survives in an academic article; and vocabulary supplied by a questionnaire is not equivalent to vocabulary independently generated by its respondents. This stratified strategy is consistent with corpus-design work emphasizing register-sensitive sampling and with history-of-emotions scholarship warning that textual evidence of emotion is socially and culturally mediated rather than transparent access to invariant internal states. ¹

The constructed Round 1B inventory contains **180 Priority Candidates—exactly 30 per anchor—and 36 Expansion Candidates**, for 216 lexical records overall. Among the 180 Priority records, 142 are R1 Core, 31 R2 Contextual, and seven R3 Peripheral/Ambiguous; none of the negative probes were used merely to fill the 30-candidate quotas. There are 156 Strong-confidence and 24 Moderate-confidence Priority records. The primary-voice assignments are V1 Scientific 71, V2 Institutional 49, V3 Mediated Public 18, V4 Organised Civic four, and V5 Direct Public 38. These are **inventory-construction statistics, not historical frequencies or measures of discourse prevalence**.

The most consequential substantive result is not a simple chronology from “no fear” to “fear.” Instead, the evidence reveals **different documentary routes by which temperature acquires social and affective significance**. In 1842, expanded worker testimony and personal correspondence reveal suffering, suffocation, fainting, complaints and heat-related bodily experience alongside meteorological measurement. In 1938, Callendar’s CO₂–temperature causal vocabulary can be placed beside vivid contemporary heat-event journalism, but no robust evidence was found tying *fear, worry, or anxiety* directly to Callendar’s anthropogenic-warming configuration. In 1988, institutional and mediated discourse becomes strongly threat-, crisis-, security-, and consequence-oriented, while exact direct-lay affect remains poorly reconstructed. In 2006–2007, mediated communication can explicitly command audiences to be “worried,” while surveys independently make *worry* an elicited response category. In 2015, temperature is conspicuously formalised as a governance threshold through 2°C/1.5°C language while “common concern” operates as a legal-diplomatic formula rather than evidence of personal emotion. By 2022, affect vocabulary is dense—but the provenance distinction becomes more, not less, important: *climate anxiety* and *eco-anxiety* are robust researcher constructs, whereas *worried, afraid, anxious, hopeless*, and related forms are separately evidenced through survey instruments and participant endorsement. ²

A major research-integrity exception must be recorded. The named mandatory seed report, **“Fear of Temperature: Historical Lexical Discovery and Anchor Validation,”** and its exact 180-record Round 1 inventory were not retrievable from the current-conversation files or from the connected Drive searches available in this session. The connected material did contain the project’s topic-definition

document, confirming the “Fear of Temperature” thesis framing and shared-cultural-heritage proposition, but not the requested Round 1 candidate table. ³ Consequently, it would be methodologically improper to fabricate a 180-row Retain/Upgrade/Downgrade mapping. Round 1B therefore does two things instead: it re-examines **all 37 Round-1 terms/probes explicitly recoverable from the present specification**, and it marks the unrecoverable exact Round-1 record identities as unresolved rather than pretending they were available. This is the one requested stopping-rule condition that cannot honestly be certified as complete.

The full structured data package contains the complete candidate schema, evidence entries, coverage audit, researcher/public audit, negative register, source registry, inventory statistics, and Round 2 shortlist:

[Download the complete Round 1B research workbook](#)

[Download the 180 Priority Candidate CSV](#)

[Download the source-coverage matrix](#)

[Download the negative/failed-evidence register](#)

[Download the source registry](#)

[Download the machine-readable JSON inventory](#)

Controlled Analytical Framework and Source-Coverage Audit

Round 1B retains the four-layer lexical architecture but prevents it from doing work that belongs to speaker analysis.

Dimension	Controlled value	Operational meaning
Layer	A — Temperature / Physical	Temperature, heat, warming, heat waves, thermal measurement or directly temperature-related phenomena
Layer	B — Climate / Atmospheric / Causal	Climate, CO ₂ , greenhouse mechanisms, atmospheric explanation, human influence or causal framing
Layer	C — Affect	Explicit concern, worry, fear, anxiety, distress, hopelessness or another evaluative/affective expression
Layer	D — Threat / Risk / Harm	Danger, threat, risk, crisis, emergency, damage, vulnerability, mortality or adverse consequence
Voice	V1 Scientific / Research	Scientist, researcher, academic, assessment author or expert
Voice	V2 Institutional / Governance	Government, UN/international body, treaty or official agency
Voice	V3 Mediated Public	Journalist, editor, documentary/public communicator or public intellectual

Dimension	Controlled value	Operational meaning
Voice	V4 Organised Civic / Advocacy	NGO, campaign, environmental organisation, movement or civic association
Voice	V5 Direct Public / Lay	Resident, worker, survey participant, witness, diarist, correspondent or other non-expert individual/group
Mode	E1 Direct / Spontaneous	Speaker independently generates wording
Mode	E2 Quoted / Mediated	Speaker's wording survives through an intermediary
Mode	E3 Elicited	Interview, survey, questionnaire or focus-group evidence
Mode	E4 Researcher-Coded	Analytical/researcher category that participants need not have used
Mode	E5 Not Applicable	Distinction does not materially apply
Relevance	R1 Core	Strongly evidenced and directly important
Relevance	R2 Contextual	Necessary to reconstruct historical lexical configuration
Relevance	R3 Peripheral / Ambiguous	Genuine attestation but polysemous, weakly connected or highly contextual
Relevance	R4 Negative Control	Actively investigated but not adequately validated

The practical effect is especially important for mixed documents. The 1842 sanitary report, for example, is an institutional/government genre but contains substantively lay worker testimony; therefore testimony such as *excessive heat*, *most suffocating*, *faint away*, and *complain of the heat* is coded V5/E3 rather than being absorbed into V2 merely because Chadwick's inquiry preserved it. Similarly, a modern academic article can contain V5 interview material while *eco-anxiety* remains a V1/E4 researcher label. This voice/genre separation is one of the largest methodological improvements over an unstratified lexical inventory. ⁴

Cross-anchor source coverage

"Strong," "Moderate," "Thin," and "Not located" below describe **recoverability and evidential quality**, not the historical importance of the voice.

Anchor	Voice	Genres actively searched	Round 1B result	Coverage
1842	V1	Observatory manuscripts; expedition meteorology; notebooks; scientific correspondence	Colaba, Ross, Leichhardt provide extensive observational vocabulary	Strong
1842	V2	Parliamentary sanitary inquiry; official observational infrastructure	Strong environmental-health context, including embedded testimony	Moderate

Anchor	Voice	Genres actively searched	Round 1B result	Coverage
1842	V3	Newspapers; reprints; public-health reporting	Heat and mortality framing found, but original-source chains can be incomplete	Moderate
1842	V4	Associations; organised civic/environmental material	No defensible heat/climate-specific organised advocacy corpus located	Not located
1842	V5	Worker testimony; personal correspondence	Chadwick testimony and personal letters materially expand lived-heat evidence	Strong
1938	V1	Scientific article; scientific reception	Callendar's causal-science lexicon exceptionally well documented	Strong
1938	V2	Official/governmental or meteorological discussion	No comparably rich climate-governance lexicon retrieved	Thin
1938	V3	Newspapers; periodicals; popular weather reporting	Strong event-level heat/drought vocabulary	Moderate
1938	V4	Civic/environmental organisations	No defensible climate/heat advocacy corpus located	Not located
1938	V5	Letters; diaries; public reaction; testimony	No strong exact-1938 lay reception of Callendar located	Thin
1988	V1	Papers; expert congressional testimony	Strong warming, attribution and extreme-event vocabulary	Strong
1988	V2	International conference/declaration; governance	Dense threat, security, crisis and consequence framing	Strong
1988	V3	Front-page journalism; commentary; scientific editorial	Strong translation of science into public-facing warning	Strong
1988	V4	NGOs; conference participation; movement material	Standalone voice-specific evidence remains under-sampled	Thin
1988	V5	Surveys; letters; citizen reaction	No sufficiently strong exact-1988 direct-public corpus located	Not located
2006–2007	V1	IPCC assessment	Mature physical/climate-impact terminology	Strong
2006–2007	V2	Economic review; official policy framing	Strong risk, cost, threat and urgency vocabulary	Strong
2006–2007	V3	Magazines; journalism; high-profile communication	Explicit affective and high-intensity public framing	Strong

Anchor	Voice	Genres actively searched	Round 1B result	Coverage
2006–2007	V4	NGO submissions and campaigns	Danger/warning vocabulary demonstrable but less comprehensively sampled	Moderate
2006–2007	V5	National and international surveys	Substantial direct-public evidence, predominantly E3	Strong
2015	V1	Climate science and research surrounding COP21	Actively searched but not prioritised heavily inside the governance-centred 30	Moderate
2015	V2	Treaty and official UN process	Exceptionally rich threshold/risk/vulnerability vocabulary	Strong
2015	V3	Journalism; public communication; public art	Broadly recoverable, though secondary to V2/V4/V5 in Priority selection	Moderate
2015	V4	COP21 mobilisation, campaign and protest material	Justice, limits, red-line and mobilisation vocabulary recoverable	Strong
2015	V5	Global/U.S. surveys; quoted demonstrators	Strong E3 survey evidence plus some E2 quotations	Strong
2022	V1	IPCC; psychology; climate-emotions research	Very dense researcher and assessment terminology	Strong
2022	V2	International assessment/official risk communication	Strong heat, mortality and mental-health-risk vocabulary	Strong
2022	V3	Heat reporting; climate media; research communication	Abundant, deliberately down-selected to prevent contemporary-source dominance	Moderate
2022	V4	Climate protest/activism	Crisis/emergency vocabulary clearly present, provenance uneven by item	Moderate
2022	V5	International/U.S. surveys; interviews	Strong evidence, but dominated by elicitation rather than E1 spontaneous text	Strong

The audit is itself a substantive result. **The empty cells differ historically.** V4 is not artificially filled for 1842 or 1938; V5 is not inferred for 1988 from newspaper prominence; and the apparent abundance of 2022 public-affect language is qualified by the fact that much of it enters the record through survey instrumentation. Those distinctions must remain visible in any subsequent corpus comparison.

Anchor-Level Expanded Lexical Inventories

The compact tables below show all 180 Priority Candidates. Bracketed codes give **Layer / primary Voice / Source ID**. Priority rank is a research-selection rank based on attestation, relevance, provenance and semantic value; **it is not a frequency rank**. The workbook supplies every additional required field—expression mode, genre, domains, historical evidence, sense, referent, directness, semantic relation, actor, affected group, audience, authority, response orientation, relevance, confidence, ambiguity, Round-1 status and Round-1B outcome.

Meteorological and environmental baseline — 1842. The Royal Society's Colaba material confirms dense thermometer- and observation-centred vocabulary in actual 1842 manuscripts, including standard, attached, dry- and wet-bulb thermometer categories. The broader Colaba sequence runs across 1841–1842. ⁵ The major expansion over a purely meteorological baseline comes from sanitary testimony and personal correspondence: labouring-space heat is described through suffocation, fainting, perspiration and complaints, while Dickens and Leichhardt supply personal/environmental wording. ⁶ Newspaper/public-health material additionally places heat beside mortality and possible public “alarm,” although that particular affective construction is negated or potential rather than evidence that a public actually reported being alarmed. ⁷

Rank	Candidate · Layer/ Voice · Source	Rank	Candidate · Layer/ Voice · Source	Rank	Candidate · Layer/Voice · Source
1	temperature · A/V1 · S1842-RS-COLABA	11	wet bulb thermometer · A/V1 · S1842-RS-COLABA	21	heat of the irons · A/V5 · S1842-CHADWICK
2	mean temperature · A/V1 · S1842-ROSS	12	meteorological observations · B/V1 · S1842-RS-COLABA	22	most suffocating · D/V5 · S1842-CHADWICK
3	temperature of the air in shade · A/V1 · S1842-ROSS	13	meteorological journal · B/V1 · S1842- ROSS	23	faint away · D/V5 · S1842-CHADWICK
4	maximum temperature · A/V1 · S1842-ROSS	14	weather · B/V1 · S1842-LEICHHARDT-L	24	perspiration · D/V5 · S1842-CHADWICK
5	minimum temperature · A/V1 · S1842-ROSS	15	climate · B/V1 · S1842- LEICHHARDT-L	25	depressing effect · C/V5 · S1842-CHADWICK
6	mean temperature of sea at surface · A/ V1 · S1842-ROSS	16	sun's heat · A/V1 · S1842-ROSS	26	complain of the heat · C/V5 · S1842- CHADWICK
7	dew point · A/V1 · S1842-LEICHHARDT- N	17	radiation of the heat · B/V1 · S1842-ROSS	27	premature heat of the weather · A/V5 · S1842- DICKENS

Rank	Candidate · Layer/ Voice · Source	Rank	Candidate · Layer/ Voice · Source	Rank	Candidate · Layer/Voice · Source
8	standard thermometer · A/V1 · S1842-RS-COLABA	18	changes of temperature · A/V2 · S1842-CHADWICK	28	suffered more from cold than from heat · D/V1 · S1842- LEICHHARDT-L
9	attached thermometer · A/V1 · S1842-RS-COLABA	19	excessive heat · D/V5 · S1842-CHADWICK	29	sickness and mortality · D/V2 · S1842- CHADWICK
10	dry bulb thermometer · A/V1 · S1842-RS-COLABA	20	heat of the men · A/V5 · S1842-CHADWICK	30	alarm to the public · C/ V3 · S1842-TIMES

The key correction is therefore not “1842 contained climate fear.” It is that **Round 1’s weak affect evidence cannot be explained solely by an absence of lived language**: once V5 sources are added, explicit bodily suffering and evaluation appear. Yet *fear of climate*, *global warming* and modern *climate anxiety* remain unvalidated. The evidence warrants a richer heat-harm configuration, not a retrospective modern climate-emotion category.

Expansion: **the heat is remarkable; 92° in the shade; 80 yesterday in the shade; very depressing effect on the energies; fatal; mortality.**

CO₂-temperature causal-science bridge — 1938. Callendar’s April 1938 article explicitly joins fuel combustion, added atmospheric carbon dioxide, radiation absorption, “sky radiation,” an estimated increase in mean temperature, meteorological-station observations, and rising world temperatures. This is unusually strong primary-source evidence for the causal/scientific lexical bridge. ⁸ Contemporary reception was not uniformly accepting, and later historical work reproduces sceptical scientific language; this should be coded as epistemic evaluation, not “fear.” ⁹

Source expansion also uncovered vivid heat-event public language independently of Callendar. The 12 December 1938 *Courier-Mail* report survives in a secondary reproduction with headlines including “Hottest Day for Ten Years” and “Heat Wave Over Queensland,” plus terms such as scorching, pitiless, drought and grass fires; because the primary page itself was not directly inspected in this pass, those records are Moderate rather than Strong. ¹⁰ A December 1938 school periodical also contains *unbearable heat* in a drought narrative, but its literary character makes it peripheral evidence rather than spontaneous lay testimony. ¹¹

Rank	Candidate · Layer/ Voice · Source	Rank	Candidate · Layer/Voice · Source	Rank	Candidate · Layer/ Voice · Source
1	artificial production of carbon dioxide · B/ V1 · S1938- CALLENDAR	11	temperature observations · A/V1 · S1938-CALLENDAR	21	heat wave · A/V3 · S1938-COURIER
2	carbon dioxide · B/V1 · S1938-CALLENDAR	12	meteorological stations · B/V1 · S1938-CALLENDAR	22	40-degree temperatures · A/ V3 · S1938- COURIER

Rank	Candidate · Layer/ Voice · Source	Rank	Candidate · Layer/Voice · Source	Rank	Candidate · Layer/ Voice · Source
3	fuel combustion · B/ V1 · S1938- CALLENDAR	13	world temperatures · A/ V1 · S1938-CALLENDAR	23	scorching westerly wind · D/V3 · S1938- COURIER
4	air · B/V1 · S1938- CALLENDAR	14	actually increased · A/V1 · S1938-CALLENDAR	24	pitiless sun · D/V3 · S1938-COURIER
5	atmosphere · B/V1 · S1938-CALLENDAR	15	average rate · A/V1 · S1938-CALLENDAR	25	hottest spell for years · A/V3 · S1938-COURIER
6	radiation absorption coefficients · B/V1 · S1938-CALLENDAR	16	small increases in mean temperature · A/V1 · S1938-WIRED- CALLENDAR	26	weather is just bearable · C/V3 · S1938-COURIER
7	water vapour · B/V1 · S1938-CALLENDAR	17	deadly glaciers · D/V1 · S1938-WIRED- CALLENDAR	27	heat records · A/V3 · S1938-COURIER
8	sky radiation · B/V1 · S1938-CALLENDAR	18	rather a coincidence · C/ V1 · S1938-SHI- RECEPTION	28	drought · D/V3 · S1938-COURIER
9	increase in mean temperature · A/V1 · S1938-CALLENDAR	19	Hottest Day for Ten Years · A/V3 · S1938-COURIER	29	grass fires · D/V3 · S1938-COURIER
10	mean temperature · A/V1 · S1938- CALLENDAR	20	Heat Wave Over Queensland · A/V3 · S1938-COURIER	30	unbearable heat · D/V3 · S1938- TORCH

The central negative result survives broader searching: **heat danger is demonstrable in 1938 public media, but direct fear/worry/anxiety about anthropogenic CO₂ warming is not thereby established.** The failure of *fear* in Callendar cannot be generalized into “society had no climatic fear,” but neither can unrelated heat-wave suffering be converted into evidence of fear of anthropogenic warming.

Expansion: **blinding sun; choking death from thirst and starvation; birds dropped dead; thermometers burst; heat-related illness** [secondary wording; weak]; **warming trend** [secondary historical locator].

Institutional climate-risk anchor — 1988. Hansen's June testimony supplies *greenhouse effect, global warming, cause-and-effect relationship, global temperature, regional heat waves, drought, extreme events*, models and observations. His central claim explicitly linked observed warming and the greenhouse effect with a high degree of confidence. ¹² The Toronto Conference then supplies a markedly different institutional register: changing atmosphere, climate warming, international/global security, harmful consequences, grave/major threats, economic/social dislocation, conflict risk and crisis. ¹³ Journalism in turn placed the scientific claim into a public-facing headline frame—“Global Warming Has Begun”—while other 1988 commentary used directly dangerous language about excess carbon dioxide trapping heat. ¹⁴

Rank	Candidate · Layer/ Voice · Source	Rank	Candidate · Layer/ Voice · Source	Rank	Candidate · Layer/Voice · Source
1	greenhouse effect · B/V1 · S1988- HANSEN	11	extreme events · D/V1 · S1988- HANSEN	21	grave threats · D/V2 · S1988-TORONTO
2	global warming · A/ V1 · S1988-HANSEN	12	global climate models · B/V1 · S1988-HANSEN	22	severe economic and social dislocation · D/V2 · S1988- TORONTO
3	earth is warmer · A/ V1 · S1988-HANSEN	13	global observations · B/V1 · S1988-HANSEN	23	risk of conflicts · D/V2 · S1988-TORONTO
4	instrumental measurements · B/ V1 · S1988-HANSEN	14	warm air · A/V1 · S1988-HANSEN	24	impending crisis · D/V2 · S1988-TORONTO
5	cause-and-effect relationship · B/V1 · S1988-HANSEN	15	changing atmosphere · B/V2 · S1988-TORONTO	25	degradation of the atmosphere · D/V2 · S1988- TORONTO
6	global temperature · A/V1 · S1988- HANSEN	16	global security · D/ V2 · S1988- TORONTO	26	atmospheric pollution · D/ V2 · S1988-TORONTO
7	regional heat waves · A/V1 · S1988- HANSEN	17	climate warming · A/V2 · S1988- TORONTO	27	greenhouse gases · B/V2 · S1988-TORONTO
8	heat wave · A/V1 · S1988-HANSEN	18	changed climatic extremes · D/V2 · S1988-TORONTO	28	Global Warming Has Begun · A/V3 · S1988-NYT
9	drought · D/V1 · S1988-HANSEN	19	major threat · D/V2 · S1988-TORONTO	29	evidence is pretty strong · C/V3 · S1988-NYT
10	heat wave/drought occurrences · D/V1 · S1988-HANSEN	20	harmful consequences · D/ V2 · S1988- TORONTO	30	too much carbon dioxide is dangerous · D/V3 · S1988- EVENINGSTANDARD

The important qualification is V5. Round 1B found much stronger **public-facing** language than institutional documents alone reveal, but “public-facing” is not “public-generated.” No sufficiently strong exact-1988 direct-public corpus was located to establish widespread self-reported *fear, worry, alarm* or *anxiety*. Treating the latter as present merely because newspapers and international conferences were alarmed would reproduce precisely the voice-confusion Round 1B was designed to prevent.

Expansion: **enemy is invisible but deadly; traps heat from the sun; global nuclear war; high degree of confidence; probability of occurrence; more frequent.**

Public communication and threat-framing bridge — 2006–2007. IPCC AR4 makes physical warming and its evidence explicit through *warming of the climate system, unequivocal*, global air/ocean temperatures, surface temperature, snow/ice melt and sea-level rise. ¹⁵ Stern's economic/governance

register instead emphasises a serious global threat, urgent response, costs, risks, damage and human consequences. ¹⁶ TIME's 3 April 2006 cover is analytically crucial because "Be Worried. Be Very Worried." is **V3 media-generated affective direction**, whereas Gallup and Yale evidence from 2007 supplies **V5/E3 responses to researcher-provided questions/categories**. ¹⁷

Rank	Candidate · Layer/Voice · Source	Rank	Candidate · Layer/Voice · Source	Rank	Candidate · Layer/Voice · Source
1	warming of the climate system · A/V1 · S0607-IPCC	11	global warming · A/V3 · S0607-TIME	21	Be Worried. Be Very Worried. · C/V3 · S0607-TIME
2	unequivocal · B/V1 · S0607-IPCC	12	serious global threat · D/V2 · S0607-STERN	22	personally worry · C/V5 · S0607-GALLUP
3	global average air and ocean temperatures · A/V1 · S0607-IPCC	13	urgent global response · D/V2 · S0607-STERN	23	worry a great deal · C/V5 · S0607-GALLUP
4	global surface temperature · A/V1 · S0607-IPCC	14	costs and risks of climate change · D/V2 · S0607-STERN	24	serious problem · D/V5 · S0607-YALE-MAR
5	temperature increase · A/V1 · S0607-IPCC	15	impacts of climate change · D/V2 · S0607-STERN	25	danger · D/V5 · S0607-YALE-MAR
6	rising global average sea level · D/V1 · S0607-IPCC	16	damage · D/V2 · S0607-STERN	26	urgent threat · D/V5 · S0607-YALE-JUL
7	widespread melting of snow and ice · D/V1 · S0607-IPCC	17	hunger · D/V2 · S0607-STERN	27	dangerous impacts · D/V5 · S0607-YALE-JUL
8	anthropogenic CO ₂ · B/V1 · S0607-IPCC	18	water shortages · D/V2 · S0607-STERN	28	drastic action · D/V5 · S0607-YALE-JUL
9	anthropogenic GHG emissions · B/V1 · S0607-IPCC	19	coastal flooding · D/V2 · S0607-STERN	29	dangerous climate change · D/V4 · S0607-CCNI
10	climate change · B/V2 · S0607-STERN	20	worst impacts · D/V2 · S0607-STERN	30	threatens serious negative effects · D/V4 · S0607-CCNI

This period therefore offers an unusually useful same-period comparison: *worried* belongs simultaneously to a public-communication strategy and to survey instrumentation, but the two attestations do different social work. One **tells** a mass readership what orientation it should adopt; the other **asks** individuals to place themselves inside a supplied worry scale. Neither is equivalent to spontaneous E1 public self-description.

Expansion: **effects already begun; human activities; media underestimates problem; very serious concern; religious duty; serious challenge of global climate change**. Cross-national polling further shows that seriousness/concern was already being elicited internationally by 2007. ¹⁸

Temperature-threshold governance anchor — 2015. The Paris Agreement makes temperature an unusually explicit governed future object. Article 2 binds the “threat of climate change” to a global-average-temperature objective “well below” 2°C and an effort toward 1.5°C relative to pre-industrial levels; elsewhere the text explicitly develops risks/impacts, vulnerability, vulnerable groups, loss and damage, extreme and slow-onset events, early warning, emergency preparedness, irreversible/permanent losses, risk assessment/management and non-economic losses. These terms were verified against the official agreement PDF, including visual inspection of the relevant pages. ¹⁹

Rank	Candidate · Layer/ Voice · Source	Rank	Candidate · Layer/ Voice · Source	Rank	Candidate · Layer/Voice · Source
1	climate change · B/ V2 · S2015-PARIS	11	climate resilience · D/ V2 · S2015-PARIS	21	comprehensive risk assessment and management · D/V2 · S2015-PARIS
2	threat of climate change · D/V2 · S2015-PARIS	12	vulnerability · D/V2 · S2015-PARIS	22	non-economic losses · D/V2 · S2015-PARIS
3	global average temperature · A/V2 · S2015-PARIS	13	vulnerable groups · D/ V2 · S2015-PARIS	23	common concern of humankind · C/V2 · S2015-PARIS
4	well below 2°C above pre-industrial levels · A/V2 · S2015- PARIS	14	loss and damage · D/ V2 · S2015-PARIS	24	climate justice · D/V2 · S2015-PARIS
5	1.5°C above pre- industrial levels · A/ V2 · S2015-PARIS	15	extreme weather events · D/V2 · S2015- PARIS	25	very worried · C/V5 · S2015-YALE
6	pre-industrial levels · B/V2 · S2015-PARIS	16	slow onset events · D/ V2 · S2015-PARIS	26	somewhat worried · C/ V5 · S2015-YALE
7	risks and impacts · D/V2 · S2015-PARIS	17	risk of loss and damage · D/V2 · S2015-PARIS	27	very serious problem · D/V5 · S2015-PEW
8	adverse impacts · D/ V2 · S2015-PARIS	18	early warning systems · D/V2 · S2015-PARIS	28	harm them personally · D/V5 · S2015-PEW
9	adverse effects of climate change · D/ V2 · S2015-PARIS	19	emergency preparedness · D/V2 · S2015-PARIS	29	drought or water shortages · D/V5 · S2015-PEW
10	greenhouse gas emissions · B/V2 · S2015-PARIS	20	irreversible and permanent loss and damage · D/V2 · S2015-PARIS	30	We are nature defending itself · C/V4 · S2015-COPPROTEST

The most important semantic pre-validation here is **common concern ≠ individual concern**. *Common concern of humankind* is a formal legal/diplomatic construction. By contrast, Yale and Pew surveys

provide elicited V5 evidence for *worried*, *serious problem*, personal harm and anticipated climate impacts. ²⁰ Organised COP21 protest also supplies a distinct V4 lexicon of climate justice, red lines, planetary defence and mobilisation, while some individual demonstrators survive through E2 quotation. ²¹

Expansion: **severe weather; long periods of unusually hot weather; red line; at risk from climate change; for the planet; future generations.**

Contemporary heat/risk/affect endpoint — 2022. Large 2022 Yale surveys provide direct but overwhelmingly **elicited** public evidence. The international survey interviewed more than 100,000 respondents across a very large cross-national sample and directly supplied worry/threat/harm questions; the December U.S. study offered emotion terms including *afraid*, *anxious*, *hopeless*, *sad*, *angry*, *frustrated*, *hopeful*, *disgusted*, *outraged* and *depressed*. Endorsement of those words is legitimate V5/E3 evidence, but their presence in the instrument is not E1 evidence that respondents independently selected those lexical categories in unconstrained discourse. ²²

Meanwhile the IPCC directly links additional warming with heatwave exposure, heat-related mortality, ill health, premature death and mental-health challenges including anxiety and stress. ²³ Psychological research separately makes *climate anxiety*, *eco-anxiety*, *psychological distress*, powerlessness, guilt and related concepts analytically visible. Rehling's semi-structured interview study, for instance, explicitly treats eco-anxiety as a research label and derives themes from interview material; Kurth and Pihkala describe eco-anxiety as a broader family of related ecological emotions rather than a single uncomplicated synonym. ²⁴

Rank	Candidate · Layer/ Voice · Source	Rank	Candidate · Layer/Voice · Source	Rank	Candidate · Layer/ Voice · Source
1	climate change · B/ V5 · S2022-YALE-INT	11	hopeful · C/V5 · S2022- YALE-US	21	guilt · C/V1 · S2022- REHLING
2	global warming · A/ V5 · S2022-YALE-US	12	disgusted · C/V5 · S2022-YALE-US	22	heatwaves · A/V1 · S2022-IPCC
3	very worried · C/V5 · S2022-YALE-INT	13	outraged · C/V5 · S2022-YALE-US	23	heat-related mortality · D/V1 · S2022-IPCC
4	somewhat worried · C/V5 · S2022-YALE- INT	14	depressed · C/V5 · S2022-YALE-US	24	mental health challenges · D/V1 · S2022-IPCC
5	afraid · C/V5 · S2022-YALE-US	15	symptoms of anxiety · C/V5 · S2022-YALE-US	25	anxiety and stress · C/ V1 · S2022-IPCC
6	anxious · C/V5 · S2022-YALE-US	16	climate anxiety · C/V1 · S2022-NATURE	26	ill health · D/V1 · S2022-IPCC
7	hopeless · C/V5 · S2022-YALE-US	17	eco-anxiety · C/V1 · S2022-REHLING	27	premature deaths · D/ V1 · S2022-IPCC
8	sad · C/V5 · S2022- YALE-US	18	psychological distress · C/V1 · S2022-ANU	28	very serious threat · D/ V5 · S2022-YALE-INT

Rank	Candidate · Layer/ Voice · Source	Rank	Candidate · Layer/Voice · Source	Rank	Candidate · Layer/ Voice · Source
9	angry · C/V5 · S2022-YALE-US	19	distress · C/V1 · S2022- REHLING	29	harm future generations · D/V5 · S2022-YALE-INT
10	frustrated · C/V5 · S2022-YALE-US	20	powerlessness · C/V1 · S2022-REHLING	30	climate crisis · D/V4 · S2022-XR

One bibliographic correction is important for the endpoint. The influential Hickman et al. global youth *climate anxiety* article was published in **December 2021**, not 2022; its associated University of Bath dataset/instrument record appeared in 2022. It should therefore be retained as near-endpoint contextual/instrument evidence rather than silently misdated as a 2022 study. ²⁵

Expansion: **climate worry; chronic uncertainty; isolation; climate emergency; chronic fear of environmental doom; fear of a degraded Earth**. *Climate emergency* is preserved primarily as advocacy/threat framing; *chronic fear of environmental doom* and *fear of a degraded Earth* belong primarily to researcher/clinical discourse rather than proving spontaneous public self-labelling. ²⁶

Cross-Voice, Researcher–Public, and Semantic-Relational Pre-Validation

Researcher versus public vocabulary audit

The strongest Round 1B correction concerns 2022 affect vocabulary, but the principle applies across the chronology.

Surface term	Status of the wording	Voice / mode	What the evidence establishes	What it does not establish
climate anxiety	Researcher construct	V1/E4	The compound is an established research/communication label by the endpoint	That surveyed publics independently call themselves “climate anxious”
eco-anxiety	Researcher/field construct	V1/E4	Researchers explicitly theorise and operationalise the category	That interviewees necessarily generated the compound themselves
psychological distress	Research variable	V1/E4	A psychological construct can be statistically related to climate worry	Participant-generated lexical form

Surface term	Status of the wording	Voice / mode	What the evidence establishes	What it does not establish
powerlessness / guilt / chronic uncertainty	Researcher thematic labels from interview analysis	V1/E4, based on V5 material	These themes are analytically derived from elicited interviews	Transcript-level proof that every surface label was participant-produced
anxious / afraid / hopeless	Instrument options and elicited endorsements	V5/E3	Respondents can endorse supplied emotion words in relation to global warming	Unprompted E1 usage of the compound <i>climate anxiety</i>
very worried / somewhat worried	Instrument categories	V5/E3	Respondents locate themselves on a supplied worry scale	That “worry” was independently selected as their preferred vocabulary
Be Worried. Be Very Worried.	Media imperative	V3/E5	A mass communicator deliberately prescribes an affective orientation	Evidence that its readers actually felt worried
common concern of humankind	Treaty/legal formula	V2/E5	Climate change is positioned as a shared object of international concern	Individual fear or anxiety
alarm to the public	Mediated public-health formulation, partly negated	V3/E5	“Alarm” existed as a possible public-response category in the report	Proof that an identifiable public actually reported alarm

This distinction is empirically supported by the combination of survey instruments, research articles and treaty/media texts rather than inferred from lexical similarity. ²⁷

Preliminary cross-voice matrix

Term / family	V1 Scientific	V2 Institutional	V3 Mediated	V4 Civic	V5 Direct public	Preliminary functional contrast
temperature — 1842	Attested: measurement	Attested: sanitary condition	Weather/ health context	Not located	Attested: embodied/ personal conditions	Measurement versus environmental exposure

Term / family	V1 Scientific	V2 Institutional	V3 Mediated	V4 Civic	V5 Direct public	Preliminary functional contrast
heat — 1842	Attested	Attested/ contextual	Attested	Not located	Strongly attested	Same root spans physical source, sensation and harm
carbon dioxide — 1938	Strongly attested	Thin	Public reception not established in core set	Not located	Not located	Predominantly scientific in recovered 1938 causal evidence
heat wave — 1938	Separate from Callendar's core causal vocabulary	Thin	Strongly attested	Not located	Thin	Public weather-event discourse must not stand in for CO ₂ reception
global warming — 1988	Technical/ causal/ observational	Embedded in governance ecology	Headline/ public frame	Thin	Not located	Public-facing uptake is demonstrable; direct-public uptake is not
threat — 1988	Impact/risk context	Strong security frame	Strong danger framing	Thin	Not located	Institutional threat ≠ lay fear
global warming — 2006-07	Scientific object	Policy object	Public issue	Civic issue	Survey object	Same surface form performs different discourse functions
worried — 2006-07	Not central	Not central	Explicit imperative	Peripheral	Explicit E3 response category	Communicator prescription versus elicited personal orientation
climate change — 2015	Scientific context	Governed threat	Media issue	Mobilising object	Survey object	Referential overlap does not imply same framing
concern — 2015	Research/ public-attitude concept	Legal "common concern"	Public-discourse term	Advocacy concern	Elicited worry/ concern	Highest-value institutional/ affective boundary

Term / family	V1 Scientific	V2 Institutional	V3 Mediated	V4 Civic	V5 Direct public	Preliminary functional contrast
climate anxiety — 2022	Strong researcher construct	Not core	Media/research communication	Used in activist/mental-health ecology	Direct self-label not established by core surveys	Research/public mismatch
anxious — 2022	Mental-health assessment	IPCC assessment ecology	Media discourse	Possible	Strong E3 endorsement	Better direct-public evidence than assuming compound self-label
crisis — 2022	Contextual in research	Risk/governance ecology	Strong	Strong	E1 public self-use remains under-reconstructed	Intensity/frame varies with voice

Provisional lexical families

The expanded inventory supports at least the following families for Round 2. They are **families of potentially related forms, not synonym sets**.

Family	Representative surface forms	Anchors	Major ambiguity	Semantic priority
Temperature	temperature; mean temperature; global temperature; global average temperature; 2°C; 1.5°C	1842–2015	Local observation, global aggregation and normative threshold	High
Heat	heat; excessive heat; heat wave; heatwaves; unusually hot weather	1842–2022	Sensation/source, occupational exposure, event class, climate hazard	High
Warming	increase in mean temperature; climate warming; global warming; warming of climate system	1938–2022	Observation, process, attribution and issue label	High
Climate	climate; climate change; changing atmosphere; climate system	1842–2022	Local climatic character versus global anthropogenic problem-object	High

Family	Representative surface forms	Anchors	Major ambiguity	Semantic priority
Carbon/ greenhouse causality	carbon dioxide; artificial production; fuel combustion; greenhouse effect; greenhouse gases; anthropogenic emissions	1938–2015	Mechanistic, attributional and policy functions	High
Concern	alarm; concern; common concern	1842–2015	Individual affect versus institutional/legal stance	High
Worry	Be Worried; personally worry; very worried; climate worry	2006–2022	Imperative versus instrument category versus researcher variable	High
Fear	afraid; fear; fear of a degraded Earth	2022 plus earlier failed probes	Verb/adjective, analytical category and direct self-description	High
Anxiety	anxious; symptoms of anxiety; climate anxiety; eco-anxiety	2022	Emotion, symptom measure, compound construct	High
Distress	depressing effect; distress; psychological distress	1842/2022	Physiological/energetic nineteenth-century sense versus psychological construct	High
Danger/threat	dangerous; major threat; grave threats; serious global threat; threat of climate change	1988–2022	Security, economic, legal and personal objects	High
Risk	risk of conflicts; costs and risks; risk of loss and damage; risk assessment	1988–2015	Probability, economics and governance	High
Crisis	impending crisis; climate crisis; in crisis	1988/2022	Generic crisis versus compound issue label	High
Emergency	emergency preparedness; climate emergency	2015/2022	Response category versus declaration that climate itself constitutes emergency	High

Family	Representative surface forms	Anchors	Major ambiguity	Semantic priority
Harm/ consequences	mortality; harmful consequences; damage; loss and damage; harm future generations; heat-related mortality	1842–2022	Bodily, economic, legal, ecological and intergenerational harm	High

A particularly important anti-presentist test is the **distress/depression family**. The 1842 worker's "depressing effect" concerns bodily energy and oppressive working conditions; its spelling resemblance to modern *depressed*, *psychological distress* or climate-related mental-health constructs cannot itself establish semantic continuity. Conversely, the absence of modern compounds in nineteenth-century sources cannot establish absence of environmental emotion. This is precisely the type of case requiring contextual semantic analysis rather than Ngram-style string matching.

Cross-anchor recurrence register

Recurring form/family	Anchors	Provisional issue	Round 2 priority
temperature	1842, 1938, 1988, 2006–07, 2015	Measurement → aggregation → governance-threshold functions require testing	High
heat / heat wave	1842, 1938, 1988, 2015, 2022	Bodily/occupational heat, meteorological event and climate hazard may differ materially	High
climate	1842 onward	Extreme false-continuity risk between local climatic character and global anthropogenic problem	High
warming	1938 onward	Quantified increase, causal process, public headline and survey object	High
worry / worried	2006–07, 2015, 2022	Media imperative, instrument category and research variable	High
threat	1988, 2006–07, 2015, 2022	Security, economic, legal and survey functions	High
risk	1988, 2006–07, 2015, 2022	Technical probability versus economic and governance categories	High
concern	2006–07, 2015, with earlier alarm analogue	Diplomatic/legal stance versus personal orientation	High
anxiety	2022	Adjective, symptom and compound researcher construct	High
crisis	1988, 2022	Generic impending crisis versus compound climate-crisis framing	High

Recurring form/family	Anchors	Provisional issue	Round 2 priority
damage / harm	1842 onward	Changing affected subjects and temporal horizons	Medium

No row above establishes a trajectory such as “concern evolved into anxiety” or “risk became crisis.” It identifies **comparanda for testing**.

Social-relational pre-validation

The strongest actor/object relations also vary systematically without requiring a social-network model. In 1842 scientific authority observes temperature, while workers describe heat acting on bodies inside workplaces and correspondents describe travel/weather experience. In 1938 Callendar's actor-position is explanatory: human fuel combustion → atmospheric CO₂ → radiation/temperature. In 1988 Hansen's scientific warning is directed toward lawmakers while Toronto governance discourse turns climatic/atmospheric change into an international-security object. In 2006–2007 institutional experts direct governments toward risk response, journalists direct audiences toward concern, and survey instruments position respondents as evaluators. Paris positions Parties and international governance in relation to future temperature thresholds, vulnerable groups, loss and damage. By 2022 scientific assessments describe populations exposed to heat and mental-health harms while researchers simultaneously construct analytical categories through which participant reactions are classified. ²⁸

Those are relational observations supported by explicit source roles. They are **not** claims about causal social networks, public persuasion effects, or lexical evolution.

Negative Evidence, Round-1 Reconciliation, Statistics, and Round-2 Package

Negative and failed-evidence register

Negative evidence is retained because omitting it would bias the later semantic study toward successful modern-looking terms.

Anchor	Probe	Round 1B result	Interpretation
1842	global warming	Remain Negative	Not validated as 1842 surface vocabulary
1842	climate change in modern anthropogenic sense	Remain Negative	<i>Climate</i> is attested, but the modern compound/problem configuration is not
1842	fear of heat / climate fear	Unresolved	Heat suffering/evaluation exists; direct climate-related fear was not established
1842	climate anxiety	Remain Negative	Deliberate anti-anachronism control

Anchor	Probe	Round 1B result	Interpretation
1938	fear tied to anthropogenic warming	Remain Negative	Wider public heat language did not validate this connection
1938	worry / anxiety / alarm tied to Callendar configuration	Remain Negative	No robust evidence found
1938	greenhouse effect as Callendar's principal recovered surface label	Unresolved	Not established from the retrieved title/abstract; no claim of full-paper absence
1988	direct-public fear/anxiety	Unresolved	Institutional/media intensity cannot substitute for V5 evidence
1988	standalone V4 NGO vocabulary	Unresolved	Multi-actor conference participation is insufficient for speaker attribution
2006–07	climate anxiety	Remain Negative as core period public label	<i>Worry</i> is better evidenced; retrospective relabelling would be presentist
2006–07	eco-anxiety	Remain Negative as core period public label	Not validated in the priority evidence
2015	common concern = personal emotion	Reclassify	Legal/diplomatic formula
2015	climate anxiety as dominant public term	Remain Negative	Survey/public evidence favours worry/concern/seriousness
2022	climate anxiety = participant term	Reclassify	Strong V1 construct; direct V5 compound self-labelling not established by core surveys
2022	eco-anxiety = spontaneous public self-label	Unresolved	Transcript-level evidence is required
2022	climate emergency = individual affect	Reclassify	Primarily D-layer threat/advocacy/governance framing

These should be interpreted as **findings about the evidence retrieved**, never as proof that no individual anywhere used such wording.

Round-1 seed reconciliation

Because the actual Round-1 180-row table is unavailable, only the 37 seed items/probes explicitly recoverable from the supplied Round-1B specification can be responsibly given individual outcomes. The results are:

Outcome	Prompt-explicit seed/probe records	Main pattern
Retain	16	Core scientific, threshold and risk terms remain well evidenced
Upgrade	7	Broader voice searching strengthens several heat, danger, worry and public-affect terms
Reclassify	5	Especially institutional/public and researcher/participant distinctions
Remain Negative	5	Mostly anachronistic or unvalidated affect probes
Unresolved	4	Evidence insufficient or exact Round-1 provenance inaccessible
Exact Round-1 records unavailable for rowwise reconciliation	180 expected seed records	A complete 180-row change log would be fabricated without the seed inventory

Notable prompt-level outcomes include: **1842 heat** → **Upgrade** because direct worker/personal evidence broadens the source base; **1938 fear** → **Remain Negative**; **1988 threat** → **Upgrade** through Toronto's security framing; **2006–07 concern/worry evidence** → **Upgrade** through direct surveys; **2015 common concern** → **Reclassify** as legal/diplomatic rather than affective; **2022 climate anxiety and eco-anxiety** → **Reclassify** as strong research vocabulary requiring separate participant evidence; **2022 psychological distress** → **Reclassify** as a research variable; and **afraid/worry/hopeless forms** → **Upgrade** because instrument-level public evidence is directly recoverable.

Inventory statistics

Anchor	Priority	Expansion	A	B	C	D	V1	V2	V3	V4	V5	Strong / Moderate
1842	30	6	16	5	3	6	18	2	1	0	9	26 / 4
1938	30	6	13	9	2	6	18	0	12	0	0	15 / 15
1988	30	6	8	7	1	14	14	13	3	0	0	28 / 2
2006–07	30	6	5	4	3	18	9	10	2	2	7	28 / 2
2015	30	6	3	3	4	20	0	24	0	1	5	30 / 0
2022	30	6	2	1	20	7	12	0	0	1	17	29 / 1
Total	180	36	47	29	33	71	71	49	18	4	38	156 / 24

The striking layer changes in this table—A-heavy in 1842/1938, D-heavy in 1988–2015, C-heavy in 2022—**must not be read as historical word-frequency curves**. They partly reflect the historical source structures that Round 1B is explicitly documenting: 2015's Priority inventory is heavily treaty/

governance-based; 2022's is deliberately rich in psychology/survey material. The source-coverage audit must accompany these counts in every subsequent analysis.

Pre-validation package for semantic Round 2

Priority family	Terms	Anchors / voices	Preliminary Round-2 question	Why deep semantic work is necessary
Temperature / threshold	temperature; mean/global temperature; 2°C; 1.5°C	1842–2015; V1/V2/V5	When does a measured environmental quantity also function as a governed future threshold?	Links measurement to social meaning without presuming a one-way evolution
Heat	heat; excessive heat; heat wave; heatwaves	1842–2022; V1/V2/V3/V5	How do sensation, occupational exposure, event classification and climate hazard differ?	Identical roots conceal major referential differences
Climate	climate; climate change; changing atmosphere	1842–2022; all voices	What scale and causal presuppositions accompany <i>climate</i> at each anchor?	Highest presentism/false-continuity risk
Warming	temperature increase; global warming; climate warming	1938–2022; V1/V2/V3/V5	When is warming a measurement, process, causal claim, headline or survey object?	Central bridge between science and public discourse
Carbon/ greenhouse	artificial CO ₂ ; greenhouse effect; GHGs; anthropogenic emissions	1938–2007; V1/V2/V3	How do mechanism, attribution and public explanation differ?	Causal vocabulary changes speaker and communicative function
Concern/ alarm	alarm; concern; common concern	1842–2015; V2/V3/V5	Which uses are affective and which are institutional/ evaluative?	Central emotion-versus-stance boundary
Worry	Be Worried; personally worry; very worried; climate worry	2006–2022; V1/V3/V5	How do command, elicited response and researcher variable differ?	Ideal same-root cross-voice case

Priority family	Terms	Anchors / voices	Preliminary Round-2 question	Why deep semantic work is necessary
Fear/afraid	fear; afraid; fear of a degraded Earth	2022 plus earlier probes; V1/V5	Where is fear participant-generated, supplied or analytical?	Directly tests thesis title without projecting backward
Anxiety	anxious; symptoms of anxiety; climate anxiety; eco-anxiety	2022; V1/V5	How do adjective, symptom scale and compound construct relate?	Highest-priority researcher/public mismatch
Threat	major/grave/serious threat; threat of climate change	1988–2022; V2–V5	What happens to the object of threat across security, economics, law and surveys?	Threat must not be used as proxy for fear
Risk	conflict risk; economic risks; loss-and-damage risk	1988–2015; V1/V2/V5	When is risk probabilistic, economic or governance-based?	Separates technical risk from affect
Crisis/emergency	impending crisis; climate crisis; emergency preparedness; climate emergency	1988–2022; V2/V3/V4	Are these severity predicates, formal response categories or lexicalised issue labels?	Potential substitution/intensification is a hypothesis only
Distress/depression	depressing effect; distress; psychological distress; depressed	1842/2022; V1/V5	Are superficially related forms historically commensurable?	Exceptional false-continuity test
Harm/loss	mortality; consequences; damage; loss and damage; heat mortality	1842–2022; V1/V2/V3/V5	Who or what becomes represented as the affected subject?	Connects temperature to bodies, economies, ecosystems and generations

Round 2 should therefore prioritize **contextual concordance and source-level sense annotation before any aggregation**. Especially high-value comparisons are same-form/different-voice cases and researcher/instrument/participant mismatches. A semantic network, embeddings, sentiment model or frequency curve should not precede this source-level disambiguation.

References and Research Artifacts

The bibliography below separates sources by their primary research function. Where access was through a later reproduction rather than the original archival page, that limitation is stated rather than silently upgrading provenance. The workbook's **Source_Registry** sheet contains 51 source records with source IDs and metadata notes.

Primary historical sources.

S1842-RS-COLABA. Royal Society Archives, *Meteorological observations made at Colaba Observatory, Bombay [Mumbai], India*, MA/346/2, 13 January–30 April 1842. The catalogue describes hourly and half-hourly observations using barometer/attached thermometer, standard thermometer and dry/wet-bulb observations. ²⁹

S1842-RS-SERIES. Royal Society Archives, *Meteorological, magnetic, and astronomical observations made at Colaba Observatory, Bombay [Mumbai], India*, MA/346, 1 March 1841–22 September 1842. ³⁰

S1842-ROSS. James Clark Ross, *Voyage of Discovery and Research in the Southern and Antarctic Regions*, vol. 2, meteorological tables recording observations made in 1842. The observation date and later publication date are kept separate. ³¹

S1842-LEICHHARDT-L. Ludwig Leichhardt to Friedrich August Schmalfuß, Sydney, 23 March 1842, transcription via Environment & Society Portal. ³²

S1842-LEICHHARDT-N. Ludwig Leichhardt notebook, notes, manuscripts and letters, 1842–1847, State Library of New South Wales transcription. ³³

S1938-CALLENDAR. G. S. Callendar, "The artificial production of carbon dioxide and its influence on temperature," *Quarterly Journal of the Royal Meteorological Society* 64 (1938): 223–240, DOI 10.1002/qj.49706427503. ⁸

S1988-HANSEN-JGR. James Hansen et al., "Global climate changes as forecast by Goddard Institute for Space Studies three-dimensional model," *Journal of Geophysical Research* (1988); NASA/GISS bibliographic record used in source validation. ³⁴

S1988-NATURE. Nature editorial, "When to cherish white elephants," *Nature* 332 (10 March 1988): 97, DOI 10.1038/332097a0. ³⁵

Scholarly secondary and research sources.

S1938-SHI-RECEPTION. Science History Institute, "How Two Outsider Scientists Saw Inside Climate Change," historical reconstruction of Callendar and contemporary reception. Used to locate reception vocabulary, not as 1938 surface prose in its own right. ³⁶

S1938-WIRED-CALLENDAR. *Wired*, "Meet the Amateur Scientist Who Discovered Climate Change" (2018), historical reconstruction reproducing attributed Callendar wording. ³⁷

S2022-REHLING. Joseph T. C. Rehling, "Conceptualising eco-anxiety using an existential framework," *South African Journal of Psychology* 52(4), first published online 17 October 2022, DOI 10.1177/00812463221130898. ³⁸

S2022-KURTH. Charlie Kurth and Panu Pihkala, "Eco-anxiety: What it is and why it matters," *Frontiers in Psychology* 13 (2022): 981814, DOI 10.3389/fpsyg.2022.981814. ³⁹

S2022-NATURE. Bronwyn Wake, "Understanding eco-anxiety," *Nature Climate Change* 12 (2022): 884, DOI 10.1038/s41558-022-01497-0. ⁴⁰

S2022-ANU. Sonia L. Curll et al., "Nature Connectedness in the Climate Change Context," *Translational Issues in Psychological Science* (2022), DOI 10.1037/tps0000329. ⁴¹

S2022-HICKMAN. Caroline Hickman et al., "Climate anxiety in children and young people and their beliefs about government responses to climate change: a global survey," *The Lancet Planetary Health* 5(12) (December 2021): e863–e873, DOI 10.1016/S2542-5196(21)00278-3. This is explicitly treated as a 2021 near-endpoint source, not misdated as 2022. ⁴²

S2022-FOSTER. Stephen J. Foster, "Eco-anxiety in everyday life: facing the anxiety and fear of a degraded Earth in analytic work," *Journal of Analytical Psychology* (2022), DOI 10.1111/1468-5922.12860. ⁴³

S2022-QMETHOD. "Eco-anxiety: A Q method analysis towards eco-anxiety attitudes in the United Arab Emirates," published online 3 November 2022, DOI 10.1080/02650533.2022.2137120. ⁴⁴

Institutional and official sources.

S1842-CHADWICK. Poor Law Commissioners / Edwin Chadwick, *Report on an Inquiry into the Sanitary Condition of the Labouring Population of Great Britain*, presented to Parliament, July 1842; especially the embedded worker evidence on workplace heat. ⁴⁵

S1988-HANSEN. James E. Hansen, testimony before the U.S. Senate Committee on Energy and Natural Resources, 23 June 1988, "The Greenhouse Effect: Impacts on Current Global Temperature and Regional Heat Waves." ¹²

S1988-TORONTO. World Conference on the Changing Atmosphere: Implications for Global Security, Toronto, 27–30 June 1988, conference statement/declaration. ¹³

S0607-IPCC. IPCC, *Climate Change 2007: Synthesis Report*, Fourth Assessment Report, Geneva, 2007. ¹⁵

S0607-STERN. Nicholas Stern, *The Economics of Climate Change: The Stern Review*; UK Government review released 30 October 2006; Cambridge University Press edition 2007. ⁴⁶

S2015-PARIS. UNFCCC, *Paris Agreement*, adopted 12 December 2015, FCCC/CP/2015/10/Add.1, annex. ¹⁹

S2022-IPCC. IPCC, *Climate Change 2022: Impacts, Adaptation and Vulnerability*, Working Group II contribution to AR6, Summary for Policymakers. ²³

Media and public-discourse sources.

S1842-TIMES. *Times and Seasons*, 1842 issues containing contemporaneous/reprinted public-health reports linking excessive heat with disease/mortality. Because some material is reprinted, original-paper provenance remains incomplete. ⁷

S1938-COURIER. *The Courier-Mail* (Brisbane), 12 December 1938, heat-wave front-page report; wording and bibliographic identification accessed through the later *Kill Your Darlings* reproduction “A State of Slogans.” Candidate confidence is therefore Moderate. ¹⁰

S1938-TORCH. *The Torch-Bearer*, Sydney Church of England Grammar School, 1 December 1938, digitised periodical. ¹¹

S1988-NYT. Philip Shabecoff, “Global Warming Has Begun, Expert Tells Senate,” *The New York Times*, 24 June 1988; headline/date corroborated in historical reconstructions of Hansen's testimony. ⁴⁷

S1988-EVENINGSTANDARD. Christopher Monckton, “Why the rain won't go away,” *Evening Standard*, 2 February 1988; accessed through a reproduced historical clipping. ⁴⁸

S0607-TIME. *TIME*, cover, “Global Warming: Be Worried. Be Very Worried.”, 3 April 2006. ⁴⁹

S2022-TIME-HEAT. *TIME*, “The U.K.'s Hottest Day Ever Sparks a Climate Culture War,” July 2022. ⁵⁰

Civic and advocacy sources.

S0607-CCNI. Countryside Council for Northern Ireland, submission to the Northern Ireland Assembly climate-change inquiry, 2007; organised civic framing of dangerous climate change and effects on people/wildlife. ⁵¹

S0607-FOE. Friends of the Earth, “Big Ask” campaign materials and contemporary 2007 reporting. ⁵²

S2015-COPPROTEST. Euronews, “Demonstrations at COP21,” 12 December 2015, preserving activist chants and quoted individual demonstrators. ⁵³

S2015-STANDINGMARCH. *TIME*, coverage of *The Standing March* public artwork during COP21, 30 November 2015. ⁵⁴

S2015-CAN. Climate Action Network, public mobilisation material around COP21, November 2015. ⁵⁵

S2022-XR. Contemporary 2022 Extinction Rebellion protest discourse, preserved through media reporting, including explicit crisis framing. ⁵⁶

Direct-public and participant sources.

S1842-DICKENS. Charles Dickens to William Charles Macready, 22 March 1842, Morgan Library & Museum, Letter 9, p. 3. ⁵⁷

S0607-YALE-MAR. Yale Environmental Poll, fielded 5–11 February 2007; Yale News release, 20 March 2007. ⁵⁸

S0607-GALLUP. Gallup Environment Poll, 11–14 March 2007; Lydia Saad, “Did Hollywood's Glare Heat Up Public Concern About Global Warming?”, 21 March 2007. ⁵⁹

S0607-YALE-JUL. Yale/Gallup/ClearVision, *American Opinions on Global Warming*, interviews 23–26 July 2007; report 31 July 2007. ⁶⁰

S0607-PEW. Pew Global Attitudes, “Global Threats: The World's Shifting Agenda,” 27 June 2007. ¹⁸

S2015-PEW. Pew Research Center, *Global Concern about Climate Change*, survey of 45,435 respondents in 40 nations, fielded 25 March–27 May 2015; report 5 November 2015. ⁶¹

S2015-YALE. Leiserowitz et al., *Climate Change in the American Mind: October 2015*, Yale Program on Climate Change Communication, report 19 November 2015. ⁶²

S2022-YALE-INT. Yale Program on Climate Change Communication / Data for Good at Meta, *International Public Opinion on Climate Change, 2022*, interviews 25 March–14 April 2022; report 29 June 2022. ⁶³

S2022-YALE-US. Yale Program on Climate Change Communication, *Climate Change in the American Mind: Beliefs & Attitudes*, December 2022. ⁶⁴

Methodological and research-data sources.

S2022-HICKMAN-DATA. University of Bath Research Data Archive, Hickman et al. dataset and survey-question template; record published 26 May 2022, DOI 10.15125/BATH-01124. This is especially important for reconstructing the distinction between article title, research construct, question wording and participant response. ⁶⁵

M-BIBER. Douglas Biber, “Representativeness in Corpus Design,” *Literary and Linguistic Computing* 8(4) (1993): 243–257. ⁶⁶

M-PLAMPER. Jan Plamper, “History of emotions: paths and challenges,” *Cuadernos de Historia Contemporánea* (2014). ⁶⁷

M-BARCLAY. Katie Barclay, “State of the Field: The History of Emotions” (2021). ⁶⁸

M-PROV. W3C, *PROV Model Primer* (2013), used as a model for keeping source/entity/activity provenance explicit. ⁶⁹

The Round 1B evidence therefore supports a substantially more defensible formulation for the next stage of the thesis: **the historical problem is not simply when people “began to fear temperature,” but how measured temperature, experienced heat, climatic causation, institutional danger, mediated warning, organised mobilisation and named affect became related—through different speakers and documentary systems—without assuming in advance that their words, meanings, emotional states or social functions were equivalent.**

¹ ⁶⁶ <https://academic.oup.com/dsh/article-abstract/8/4/243/928942>
<https://academic.oup.com/dsh/article-abstract/8/4/243/928942>

² ⁵ ²⁹ Meteorological observations made at Colaba Observatory, Bombay [Mumbai], India | The Royal Society: Science in the Making
https://makingscience.royalsociety.org/items/ma_346_2/meteorological-observations-made-at-colaba-observatory-bombay-mumbai-india?utm_source=chatgpt.com

³ DaiPan_TopicDefinition
https://docs.google.com/document/d/1fmZ4cvPIC5UQ-ty_owMRKf3uq3buZTqIutAWaNaY20I/edit?usp=drivesdk

⁴ ⁶ ⁴⁵ <https://www.gutenberg.org/cache/epub/65090/pg65090-images.html>
<https://www.gutenberg.org/cache/epub/65090/pg65090-images.html>

- 7 Times and Seasons, 15 October 1842, Page 947
https://www.josephsmithpapers.org/paper-summary/times-and-seasons-15-october-1842/5?utm_source=chatgpt.com
- 8 28 The artificial production of carbon dioxide and its influence on temperature - Callendar - 1938 - Quarterly Journal of the Royal Meteorological Society - Wiley Online Library
https://rmets.onlinelibrary.wiley.com/doi/abs/10.1002/qj.49706427503?utm_source=chatgpt.com
- 9 36 <https://www.sciencehistory.org/stories/magazine/how-two-outsider-scientists-saw-inside-climate-change/>
<https://www.sciencehistory.org/stories/magazine/how-two-outsider-scientists-saw-inside-climate-change/>
- 10 <https://www.killyourdarlings.com.au/article/a-state-of-slogans/>
<https://www.killyourdarlings.com.au/article/a-state-of-slogans/>
- 11 https://fliphtml5.com/nsoqf/nzof/The_Torchbearer_1938_12/
https://fliphtml5.com/nsoqf/nzof/The_Torchbearer_1938_12/
- 12 https://www.sealevel.info/1988_Hansen_Senate_Testimony.html
https://www.sealevel.info/1988_Hansen_Senate_Testimony.html
- 13 Implications for Global Security Conference Declaration ...
https://lpr.adb.org/resource/international-conference-changing-atmosphere-implications-global-security-conference?utm_source=chatgpt.com
- 14 47 <https://time.com/5762817/climate-change-denial-history/>
<https://time.com/5762817/climate-change-denial-history/>
- 15 <https://www.ipcc.ch/report/ar4/syr/>
<https://www.ipcc.ch/report/ar4/syr/>
- 16 46 <https://www.cambridge.org/core/books/economics-of-climate-change/summary-of-conclusions/24EE5904FA4F9BEE88B862E7FF123462>
<https://www.cambridge.org/core/books/economics-of-climate-change/summary-of-conclusions/24EE5904FA4F9BEE88B862E7FF123462>
- 17 27 49 <https://content.time.com/time/covers/europe/0%2C16641%2C20060403%2C00.html>
<https://content.time.com/time/covers/europe/0%2C16641%2C20060403%2C00.html>
- 18 <https://www.pewresearch.org/global/2007/06/27/chapter-2-global-threats-the-worlds-shifting-agenda/>
<https://www.pewresearch.org/global/2007/06/27/chapter-2-global-threats-the-worlds-shifting-agenda/>
- 19 https://unfccc.int/sites/default/files/resource/parisagreement_publication.pdf
https://unfccc.int/sites/default/files/resource/parisagreement_publication.pdf
- 20 61 <https://www.pewresearch.org/global/2015/11/05/1-concern-about-climate-change-and-its-consequences/>
<https://www.pewresearch.org/global/2015/11/05/1-concern-about-climate-change-and-its-consequences/>
- 21 53 <https://www.euronews.com/2015/12/12/demonstrations-at-cop21>
<https://www.euronews.com/2015/12/12/demonstrations-at-cop21>
- 22 63 <https://climatecommunication.yale.edu/publications/international-public-opinion-on-climate-change-2022/toc/4/>
<https://climatecommunication.yale.edu/publications/international-public-opinion-on-climate-change-2022/toc/4/>
- 23 <https://www.ipcc.ch/report/ar6/wg2/chapter/summary-for-policymakers/>
<https://www.ipcc.ch/report/ar6/wg2/chapter/summary-for-policymakers/>
- 24 38 <https://journals.sagepub.com/doi/abs/10.1177/00812463221130898>
<https://journals.sagepub.com/doi/abs/10.1177/00812463221130898>

- 25 42 <https://pubmed.ncbi.nlm.nih.gov/34895496/>
<https://pubmed.ncbi.nlm.nih.gov/34895496/>
- 26 43 <https://onlinelibrary.wiley.com/doi/full/10.1111/1468-5922.12860>
<https://onlinelibrary.wiley.com/doi/full/10.1111/1468-5922.12860>
- 30 Meteorological, magnetic, and astronomical observations made at Colaba Observatory, Bombay [Mumbai], India | The Royal Society: Science in the Making
https://makingscience.royalsociety.org/items/ma_346/meteorological-magnetic-and-astronomical-observations-made-at-colaba-observatory-bombay-mumbai-india?utm_source=chatgpt.com
- 31 [https://en.wikisource.org/wiki/](https://en.wikisource.org/wiki/A_Voyage_of_Discovery_and_Research_in_the_Southern_and_Antarctic_Regions/Volume_2/Chapter_8)
A_Voyage_of_Discovery_and_Research_in_the_Southern_and_Antarctic_Regions/Volume_2/Chapter_8
https://en.wikisource.org/wiki/A_Voyage_of_Discovery_and_Research_in_the_Southern_and_Antarctic_Regions/Volume_2/Chapter_8
- 32 <https://www.environmentandsociety.org/exhibitions/ludwig-leichhardt/letter-his-brother-law-friedrich-august-schmalfuss-sydney-23-march>
<https://www.environmentandsociety.org/exhibitions/ludwig-leichhardt/letter-his-brother-law-friedrich-august-schmalfuss-sydney-23-march>
- 33 https://archival-classic.sl.nsw.gov.au/_transcript/2017/D24507/a9499.html
https://archival-classic.sl.nsw.gov.au/_transcript/2017/D24507/a9499.html
- 34 Hansen et al. 1988: Global climate changes as forecast ...
https://www.giss.nasa.gov/pubs/abs/ha02700w.html?utm_source=chatgpt.com
- 35 <https://www.nature.com/articles/332097a0>
<https://www.nature.com/articles/332097a0>
- 37 <https://www.wired.com/story/meet-the-amateur-scientist-who-discovered-climate-change>
<https://www.wired.com/story/meet-the-amateur-scientist-who-discovered-climate-change>
- 39 <https://pubmed.ncbi.nlm.nih.gov/36211934/>
<https://pubmed.ncbi.nlm.nih.gov/36211934/>
- 40 <https://www.nature.com/articles/s41558-022-01497-0>
<https://www.nature.com/articles/s41558-022-01497-0>
- 41 <https://dspace-test.anu.edu.au/items/b23d7b01-8bd7-462d-892a-118b694551a5>
<https://dspace-test.anu.edu.au/items/b23d7b01-8bd7-462d-892a-118b694551a5>
- 44 <https://www.tandfonline.com/doi/full/10.1080/02650533.2022.2137120>
<https://www.tandfonline.com/doi/full/10.1080/02650533.2022.2137120>
- 48 <https://www.desmog.com/christopher-monckton/>
<https://www.desmog.com/christopher-monckton/>
- 50 <https://time.com/6198905/uk-heat-climate-change-politics/>
<https://time.com/6198905/uk-heat-climate-change-politics/>
- 51 <https://archive.niassembly.gov.uk/environment/2007mandate/submissions/CounNatureCon.htm>
<https://archive.niassembly.gov.uk/environment/2007mandate/submissions/CounNatureCon.htm>
- 52 <https://www.theguardian.com/media/2007/jul/04/advertising>
<https://www.theguardian.com/media/2007/jul/04/advertising>
- 54 <https://time.com/4128843/paris-standing-march-jr-aronsky/>
<https://time.com/4128843/paris-standing-march-jr-aronsky/>

- 55 <https://climatenetwork.org/2015/11/21/global-climate-marches-to-see-unprecedented-wave-of-people-calling-for-more-action-ahead-of-paris-climate-summit/>
<https://climatenetwork.org/2015/11/21/global-climate-marches-to-see-unprecedented-wave-of-people-calling-for-more-action-ahead-of-paris-climate-summit/>
- 56 <https://www.theguardian.com/environment/2022/sep/02/extinction-rebellion-activists-superglue-themselves-commons-chamber-speaker-seat>
<https://www.theguardian.com/environment/2022/sep/02/extinction-rebellion-activists-superglue-themselves-commons-chamber-speaker-seat>
- 57 <https://www.themorgan.org/collection/A-Letter-from-Charles-Dickens/46>
<https://www.themorgan.org/collection/A-Letter-from-Charles-Dickens/46>
- 58 <https://news.yale.edu/2007/03/20/sea-change-public-attitudes-toward-global-warming-emerge-climate-change-seen-big-threat-t>
<https://news.yale.edu/2007/03/20/sea-change-public-attitudes-toward-global-warming-emerge-climate-change-seen-big-threat-t>
- 59 <https://news.gallup.com/poll/26932/did-hollywoods-glare-heat-public-concern-about-global-warming.aspx>
<https://news.gallup.com/poll/26932/did-hollywoods-glare-heat-public-concern-about-global-warming.aspx>
- 60 <https://climatecommunication.yale.edu/publications/american-opinions-on-global-warming/>
<https://climatecommunication.yale.edu/publications/american-opinions-on-global-warming/>
- 62 <https://climatecommunication.yale.edu/publications/more-americans-perceive-harm-from-global-warming-survey-finds/toc/2/>
<https://climatecommunication.yale.edu/publications/more-americans-perceive-harm-from-global-warming-survey-finds/toc/2/>
- 64 <https://climatecommunication.yale.edu/publications/climate-change-in-the-american-mind-beliefs-attitudes-december-2022/toc/2/>
<https://climatecommunication.yale.edu/publications/climate-change-in-the-american-mind-beliefs-attitudes-december-2022/toc/2/>
- 65 <https://researchdata.bath.ac.uk/1124/>
<https://researchdata.bath.ac.uk/1124/>
- 67 <https://revistas.ucm.es/index.php/CHCO/en/article/view/46680>
<https://revistas.ucm.es/index.php/CHCO/en/article/view/46680>
- 68 <https://onlinelibrary.wiley.com/doi/full/10.1111%2F1468-229X.13171>
<https://onlinelibrary.wiley.com/doi/full/10.1111%2F1468-229X.13171>
- 69 <https://www.w3.org/TR/prov-primer/>
<https://www.w3.org/TR/prov-primer/>